

CHEK CHEE HIM

018-407 4925 | <https://github.com/thebalanceder> |

www.linkedin.com/in/chek-chee-him

EDUCATION

Universiti Teknologi Malaysia (UTM) (Sep 2024 - Present)

CGPA: 4.0 | Degree in *Bachelor of Artificial Intelligence (Hons)* | Expected Graduation: 2027

Matriculation Pahang (KMPH) (July 2023 - May 2024)

CGPA: 4.0 | Matriculation in Science (Physics)

LEADERSHIP EXPERIENCE

Program Director of Drone Workshop: Explore, Fly and Discover (Nov 2025 - Present)

Drone Technology Class, UTM

- Leading the planning and curriculum design of a hands-on drone education workshop.

EY Data Challenge External Ambassador 2026 (Sep 2025 - Feb 2025)

Ernst & Young (EY)

- Representing the EY Data Challenge program within the student and youth community.

Department Head – Research & Development (R&D) (Dec 2024 – Present)

Society of Artificial Intelligence (ALICE), UTM

- Overseeing AI-related competitions and innovation initiatives

Chairman – Intelligence Game (Dec 2023 – May 2024)

Matriculation of Pahang

- Organized strategy-based gaming competition to enhance problem-solving skills

Research & Publications

Paper Submitted (Under Review)

“Nested Dual Gradient Boosting Model for High-Resolution Urban Heat Prediction in Dense Cities” - Malaysian Journal of Computer Science (March 2026)

- Proposed a dual-stacking gradient boosting model for accurate Urban Heat Island (UHI) index prediction.

“Enhancing Bitcoin Return Predictability through Interpretable Macro-Technical Ensembles across Market Regimes” - Springer Nature SNAPP : Computational Economics (Sep 2025)

- Interpretable machine learning framework that combines macroeconomic indicators and technical signals across different market regimes to improve the accuracy and transparency of predicting Bitcoin returns.

SKILL

Data Science: Data preprocessing, feature selection, model selection and evaluation

Programming Languages: Python (NumPy, Pandas, TensorFlow, PyTorch), C++, C

AI & Deep Learning: ensemble methods, time series forecasting, neural networks

LLMs and APIs: langchain, langgraph, agentic cli, skills

Language

Malay: Professional Working

English: Professional Working

Mandarin: Professional Working

Selected Projects & Competitions

MyAI Future Hackathon (April 2026, on-going)

- Develop an agrotech solution that uses smart technologies to detect crop diseases early and optimize farming resources, helping smallholder farmers increase yields and reduce Malaysia's reliance on food imports.

Harvard HSIL Hackathon (April 2026, on-going)

- Delivers an intelligent chatbot and knowledge system that streamlines hospital appointments and enquiries, seamlessly supporting both doctors and patients with efficient, accessible communication.

2026 EY Open Science AI & Data Challenge (Feb 2026)

- Developed AI models to forecast river water quality for regions in South Africa.

Derriv Hackathon (Feb 2026)

- Built a fraud detection system using risk score prediction with Gluon AutoML.

CHINA-ASEAN Super League: Smart Vision for Risk Detection (Jan 2026)

- Designed a custom YOLO model for violence, fire, and fall detection.
- Contributed to a multimodal AI-based early warning system for urban and rural communities.

UTM-Infineon Innovation Launchpad Project Demonstration (Dec 2025)

- Investigated and demonstrated Wi-Fi-based control using PSoC 6 for embedded applications.

Accelerator for SLAM Algorithm (FPGA) (Nov 2025)

- Explored acceleration of SLAM algorithms using KV260 FPGA for performance optimization.

CAAUEI AI Application Challenge 2025 (Jul 2025)

- Developed and fine-tuned an object detection model for real-time embedded traffic sign recognition.

2025 EY Open Science AI & Data Challenge (Mar 2025)

- Built a custom ensemble model for Urban Heat Island (UHI) prediction in New York City.

AI Hackathon – “Concentration X” (Nov 2024)

- Developed a gaze-tracking system using OpenCV.
- Combined LSTM and exponential smoothing for attention prediction over time.
- Designed a personalized study plan based on time-series analysis.

Student Innovation Competition – “Weather Monitor” (Oct 2023)

- Built an Arduino-based weather monitoring system.
- Developed a custom LSTM model to forecast Air Quality Index (AQI)